Undergraduate Seminar II
Introduction to Commutative Algebra

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§1.1 Round 6: Valuation Rings

Let us start from a definition.

Definition 1.1 (Valuation Ring)**.** Let B be an integral domain and K its field of fraction. We say that B is a **valuation ring**[1] of K if for each $x \neq 0$ (as the element of K![2]), either

$$x \in B \text{ or } x^{-1} \in B$$

or both.

Let us go through some examples:

Example

- Our old friend $B = \mathbb{Z}$ is **not** a valuation ring of $\mathbb{Q}$.
- Let
$$B = \mathbb{Z}_{(p)} = \left\{ \frac{a}{b} \in \mathbb{Q} \;\middle|\; a, b \in \mathbb{Z},\ p \nmid b \right\}.$$
Then $K = \mathbb{Q}$. $\mathbb{Z}_{(p)}$ is a valuation ring of $\mathbb{Q}$.

Proposition 1.2 (5.18)

Let B be a ring and K its field of fraction. Then, the following are true:

i) B is a local ring[a].

ii) If B' is a ring such that $B \subseteq B' \subseteq K$, then B' is a valuation ring of K.

iii) B is integrally closed (in K)[b].

[a] 局所環
[b] B は K で整閉

Proof.

i) We recall a proposition:

> **Proposition** (Recall Proposition 1.6)
> Let B be a ring, $\mathfrak{m} \neq (1)$ be an ideal of B such that for any $x \in B - \mathfrak{m}$, x is a unit in B. Then, $\mathfrak{m}$ is the unique maximal ideal of B and B is a local ring.

[1] 付値環
[2] 日本語版では、x が K の元であることが強調されていたが、英語版では特に強調されていなかった。なので、$x \in K$ だということに最初は気かなかった。

Now, similar to the 1.6, define a **set** $\mathfrak{m}$:

$$\mathfrak{m} := \{x \in B \mid x \text{ is not a unit in } B\}.$$

Then we have

$$x \in \mathfrak{m} \Leftrightarrow x = 0 \text{ or } x^{-1} \notin B.$$

We show that $\mathfrak{m}$ is an ideal of B:

1. For each $a \in B$ and $x \in \mathfrak{m}$, if $ax \notin \mathfrak{m}$ we have $(ax)^{-1} \in B$ so that

   $$x^{-1} = a(ax)^{-1} \in B,$$

   a contradiction.

2. For each $x, y \in \mathfrak{m}$, we show that $x + y \in \mathfrak{m} - 0$.
   Consider $\frac{x}{y} \in K$. Then we have $xy^{-1} \in B$ or $x^{-1}y \in B$. If, for example $xy^{-1} \in B$, we have

   $$x + y = (1 + xy^{-1})y \in B\mathfrak{m} \subseteq \mathfrak{m}.$$

Hence $\mathfrak{m}$ is an ideal of B and therefore B is a local ring by Proposition 1.6.

ii) Clear from the definition.

iii) Let $x \in K$ be integral over B. Then there exist $x_i \in B$ such that

$$x^n + b_1 x^{n-1} + \cdots + b_n = 0.$$

Now, if $x \in B$ we are done, so let $x^{-1} \in B$. Then

$$x = -(b_1 + b_2 x^{-1} + \cdots + b_n (x^{-1})^{n-1}) \in B.$$

□

Remark — From Proposition 5.18 ii) we have

$$\text{valuation ring} \Longrightarrow \text{ local ring.}$$

The converse[a] is not true, in general.

[a] 逆

Now, let K be a field, Ω an algebraically closed field[3]. Let

$$\Sigma = \{(A, f) \mid A \subseteq K \text{ subring}, f : A \to \Omega \text{ ring homomorphism}\}.$$

We partically order[4] the set Σ as follows:

$$(A, f) \leq (A', f') \iff A \subseteq A' \text{ and } f'|_A = f.$$

For each chain $\sigma \subseteq \Sigma$, the element (K, F) is an upper bound of σ thus we can use Zorn's Lemma. Consequently[5], the set Σ has atleast one maximal element, say (B, g).

Our objective is to show that B is a valuation ring of K. We need some preparation:

Lemma 1.3 (5.19)

B is a local ring and $\mathfrak{m} = \ker g$ is its maximal ideal.

Proof. Notice that $g(B)$ is a subring of a field Ω (which is an integral domain). Thus $g(B)$ is also an integral domain. Therefore, the isomoprhism

$$B/\ker g \cong g(B)$$

implies $\ker g$ a prime ideal. Now, we can extend g to a homomorphism

$$\overline{g} : B_{\mathfrak{m}} \to \Omega$$

via $\overline{g}(b/s) = g(b)/g(s)$ for all $b \in B$ and $s \in B - \mathfrak{m}$ (since $g(s)$ will not be zero). We obtain $(B, g) \leq (B_{\mathfrak{m}}, \overline{g})$, and by the maximality of (B, g) we obtain $B = B_{\mathfrak{m}}$. Hence B is a local ring and $\mathfrak{m}$ is its maximal ideal. □

Lemma 1.4 (5.20)

Let x be a non-zero element of K. Let $B[x]$ be the subring of K generated by x over B and let $\mathfrak{m}[x]$ be the extension of $\mathfrak{m}$ in $B[x]$. Then, we have either $\mathfrak{m}[x] \neq B[x]$ or $\mathfrak{m}[x^{-1}] \neq B[x^{-1}]$.

Proof. Suppose the otherwise (i.e $\mathfrak{m}[x] = B[x]$ or $\mathfrak{m}[x^{-1}] = B[x^{-1}]$). Then since $1 \in B[x]$ and $1 \in B[x^{-1}]$, there exist $u_i, v_i \in \mathfrak{m}$ such that

$$u_0 + u_1 x + \cdots + u_m x^m = 1 \tag{1.1.1}$$

$$v_0 + v_1 x^{-1} + \cdots + v_n x^{-n} = 1, \tag{1.1.2}$$

[3] 代数的閉体 (つま, 一次以上の任意の K 係数一変数多項式が K 上に根を持つこと) [4] 半順序 [5] だから

in which we assume m, n are as small as possible. WLOG $m \geq n$. Multiply eq. (1.1.2) by x^n:

$$(1 - v_0)x^n = v_1 x^{n-1} + \cdots + v_n. \tag{1.1.3}$$

Now, since $v_0 \in \mathfrak{m}$, we claim that $1 - v_0 \notin \mathfrak{m}$. This is because if not, then $1 = (1 - v_0) + v_0 \in \mathfrak{m}$, a contradiction. Thus $1 - v_0$ is a unit in B. Therefore, eq. (1.1.3) can be written in the form

$$x^n = w_1 x^{n-1} + \cdots + w_n,$$

where $w_i \in \mathfrak{m}$. Hence we can replace x^m in eq. (1.1.1) by $w_1 x^{n-1} + \cdots + w_n$, contradicting the minimality of the exponent m. □

Now, here comes the main course[6]:

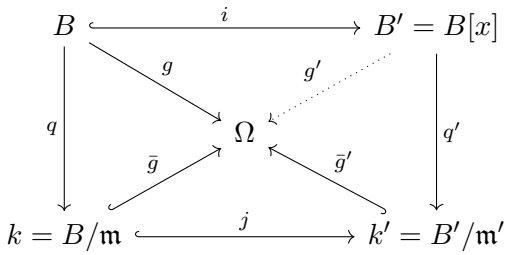
Theorem 1.5 (5.21)
Let (B, g) be a maximal element of Σ. Then B is a valuation ring of the field K.

Proof. Take a non-zero $x \in K$. Our goal it to prove $x \in B$.
Claim. There exists g' such that

$$(B, g) = (B', g').$$

Therefore, $x \in B[x] = B' = B$ implies $x \in B$.

Proof of the claim. (i.e. How do we construct g'?)
This is a diagram to help you understand the proof (it is not necessary commutative!)



Step 1. From Lemma 1.4 (5.20) $\mathfrak{m}[x]$ is not the unit ideal of ring $B' = B[x]$. Then, $\mathfrak{m}[x]$ is contained in a maximal ideal $\mathfrak{m}'$ of B' (i.e $\mathfrak{m}[x] \subseteq \mathfrak{m}' \subsetneq B'$).
Now we claim that $\mathfrak{m}' \cap B = \mathfrak{m}$. Indeed, since $\mathfrak{m}' \cap B$ is a proper ideal of B[7], we have

$$\mathfrak{m} \subseteq \mathfrak{m}' \cap B \subsetneq B.$$

Since $\mathfrak{m}$ is the maximal ideal of B, then we have $\mathfrak{m} = \mathfrak{m}' \cap B$.

[6] 主菜。ここでは、議論の核心を指す。

[7] Otherwise, we have $\mathfrak{m}' \cap B = B$, in particular $1 \in \mathfrak{m}'$, contradicting the maximality of $\mathfrak{m}'$.

Now, write $k := B/\mathfrak{m}$ and $k' = B'/\mathfrak{m}'$. m and m' are maximal ideals so k and k' are fields.

The embedding $i : B \hookrightarrow B'$ together with $\mathfrak{m} = \mathfrak{m}' \cap B$ induces an embedding

$$j : k \hookrightarrow k' \text{ via } b + \mathfrak{m} \mapsto b + \mathfrak{m}'.$$

Indeed, j is injective:

$$\ker j = \{b + \mathfrak{m} \mid b + \mathfrak{m}' \in \mathfrak{m}'\} = (B \cap \mathfrak{m}')/\mathfrak{m} = 0.$$

Step 2. Now, consider the natural map $q' : B' \to k'$ where we identify the image of x in k' as $\overline{x}$ (i.e. $\overline{x} := q'(x)$). We claim that $k' = k[\overline{x}]$.

Indeed, take $z \in k'$. Since q' is surjective, there exists $y \in B'$ such that $z = q'(y)$. We can write y as $y = a_0 + a_1 x + \cdots + a_n x^n$, where $a_i \in B$. Notice that $q'(a_i) = j(a_i + \mathfrak{m}) \in j(k) \cong k$:

$$\begin{aligned} z = q'(y) &= q'(a_0) + q'(a_1)q'(x) + \cdots + q'(a_n)q'(x)^n, \\ &= q'(a_0) + q'(a_1)\overline{x} + \cdots + q'(a_n)\overline{x}^n \in k[\overline{x}], \end{aligned}$$

hence $k' \subseteq k[\overline{x}]$.

Conversely, we know that $k \subseteq k'$ and $\overline{x} \in k'$ so $k[\overline{x}] \subseteq k'$. Moreover, x is algebraic over k and therefore k' is a finite algebraic extension of k.

Step 3. Now, the homomorphism $g : B \to k$ induces an embedding $\overline{g} : k \to \Omega$ (because $\mathfrak{m} = \ker g$). Furthermore, we extend $\overline{g} : k \to \Omega$ to an embedding $\overline{g}' : k' \to \Omega$. How does one construct such embedding?

Since Ω is algebraically closed, every non-constant polynomial in $\Omega(X)$ has root in Ω. Thus, when $\overline{x}$ is algebraic over k, the minimal polynomial is induced via $\overline{g}$ to the polynomial $\Omega(X)$, then we can choose a root in Ω as the image of $\overline{x}$. Furthermore, since $k' = k[\overline{x}]$, every element in k' can be generated by k and an element $\overline{x}$.

Let

$$f(X) = a_0 + a_1 X + \cdots + a_n X^n \in k[X]$$

be the minimal polynomial of $\overline{x}$ over k (obviously $f(\overline{x}) = 0$). Now choose $\beta \in \Omega$ such that $\overline{g}'(\overline{x}) = \beta$. Since 0 goes to 0:

$$\begin{aligned} 0 = \overline{g}'(0) &= \overline{g}'(f(\overline{x})) \\ &= \overline{g}'(a_0) + \overline{g}'(a_1)\beta + \cdots + \overline{g}'(a_n)\beta^n. \end{aligned}$$

Therefore, β is a root of a polynomial

$$f^{\overline{g}'}(X) = \overline{g}'(a_0) + \overline{g}'(a_1)X + \cdots + \overline{g}'(a_n)X^n \in \Omega[X]$$

over Ω.

Finally, the embedding $\overline{g}'$ is defined as follows:

$$\overline{g}' : k' \to \Omega \text{ where } \overline{g}'|_k = g \text{ and } \overline{x} \mapsto \beta.$$

Step 4. Finally, we define $g' : B' \to \Omega$ via $g' = \overline{g}' \circ q'$ which extends g.

Since (B, g) is already the maximal element, we left are with $B = B'$. We are done. □

Corollary 1.6 (5.22)

Let A be a subring of a field K. Then,

$$\overline{A} = \bigcap_{\substack{B \supseteq A \\ B\text{: valuation ring of } K}} B.$$

Here, $\overline{A}$ denotes the integral closure of A in K.

Proof. Let V be any valuation ring of K that contains A (i.e. $A \subseteq B$). Then $\overline{A} \subseteq \overline{B} = B$ by 5.18 iii). Since B is arbitrary, we have

$$\overline{A} \subseteq \bigcap_{\substack{B \supseteq A \\ B\text{: valuation ring of } K}} B.$$

Conversely, If $x \notin \overline{A}$, it is enough to show that $x \notin B$. Notice that x is not in the ring $A' = A[x^{-1}]$. This is because if so, then there exist $a_i \in A$ such that

$$x = a_0 + a_1 x^{-1} + \cdots + a_n x^{-n}.$$

Thus x is integral of A (i.e. $x \in \overline{A}$), a contradiction. Hence x^{-1} is not a unit in A'. Therefore, x^{-1} is contained in the maximal ideal (say $\mathfrak{m}'$) of A'.

Now, let Ω be the algebraic closure of the field $k' := A'/\mathfrak{m}'$ and define a homomoprhism $q' : A' \to k'$ and an inclusion $i : k' \to \Omega$. We then consider a pair $(A', f) \in \Sigma$ with $f := i \circ q' : A' \to \Omega$. By Zorn's lemma, we have a maximal element $(B, g) \in \Sigma$ satisfying

$$(A', f) \leq (B, g).$$

This means $A' \subseteq B$ and $g|_{A'} = f$. By Theorem 1.5 (5.21), B is a valuation ring of K. Then, since $x^{-1} \in \mathfrak{m}' \subseteq A' \subseteq B$ and $q'(x^{-1}) = 0$ in $\mathfrak{m}'$ we have

$$g(x^{-1}) = g|_{A'}(x^{-1}) = i \circ q'(x^{-1}) = 0.$$

Finally, we have $x \notin B$. This is because if $x \in B$, $g(x)$ is defined and thus

$$1 = g(1) = g(x)g(x^{-1}) = g(x) \cdot 0 = 0,$$

a contradiction (because 1 needs to be sent to 1). □

Proposition 1.7

Let $A \subseteq B$ be integral domains, B finitely generated over A. Let v be a non-zero element of B. Then there exists $u \neq 0$ in A with the following property: any homomorphism f of A into an algebraically closed field Ω such that $f(u) \neq 0$ can be extended to a homomorphism of B into Ω such that $g(v) \neq 0$.

Proof. By induction on the number of generators of B over A we reduce immediately to the case where B is generated over A by a single element x.

i) Suppose x is transcendental over A, i.e. that no non-zero polynomial with coefficients in A has x as a root. Let $v = a_0x^n + a_1x^{n-1} + \cdots + a_n$, and take $u = a_0$. Then if $f\colon A \to \Omega$ is such that $f(u) \neq 0$, there exists $\xi \in \Omega$ such that

$$f(a_0)\xi^n + f(a_1)\xi^{n-1} + \cdots + f(a_n) \neq 0,$$

because Ω is infinite. Define $g\colon B \to \Omega$ extending f by putting $g(x) = \xi$. Then $g(v) \neq 0$, as required.

ii) Now suppose x is algebraic over A (i.e. over the field of fractions of A). Then so is v^{-1}, because v is a polynomial in x. Hence we have equations of the form

$$a_0x^m + a_1x^{m-1} + \cdots + a_m = 0 \qquad (a_i \in A), \tag{1}$$

$$a_0'v^{-n} + a_1'v^{-(n-1)} + \cdots + a_n' = 0 \qquad (a_i' \in A). \tag{2}$$

Let $u = a_0a_0'$ and let $f\colon A \to \Omega$ be such that $f(u) \neq 0$. Then f can be extended, first to a homomorphism $f_1\colon A[u^{-1}] \to \Omega$ (with $f_1(u^{-1}) = f(u)^{-1}$), and then by **??** to a homomorphism $h\colon C \to \Omega$, where C is a valuation ring containing $A[u^{-1}]$. From (1), x is integral over $A[u^{-1}]$, hence by **??** $x \in C$, so that C contains B, and in particular $v \in C$. On the other hand, from (2), v^{-1} is integral over $A[u^{-1}]$, and therefore by **??** again is in C. Therefore v is a unit in C, and hence $h(v) \neq 0$. Now take g to be the restriction of h to B.

□

Corollary 1.8 (5.24)

Let k be a field and B a finitely generated k-algebra. If B is a field then it is a finite algebraic extension of k.

Proof. Take $A = k$, $v = 1$ and Ω the algebraic closure of k. □

corollary 1.8 is one form of Hilbert's Nullstellensatz. For another proof, see **??**.